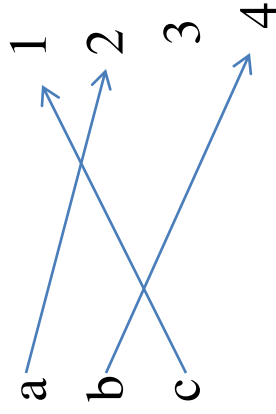


Background on functions

- A function is defined by two sets X and Y and a rule f which assigns to each element of X to an element of Y . It is denoted by $f : X \rightarrow Y$
- X : *domain* of the function f
- Y : *co-domain (range)* of the function f
- The *image* y in Y of an element x in X is denoted by $y = f(x)$
- For a function f from set X to set Y , if y in Y then a *pre-image* of y is an element x in X for which $f(x) = y$
- The set of all elements in Y which have at least one pre-image is called the image of f , denoted by $Im(f)$

- *Example:*

Consider $X = \{a, b, c\}$ and $Y = \{1, 2, 3, 4\}$, $f: X \rightarrow Y$ is defined as $f(a) = 2$, $f(b) = 4$, $f(c) = 1$. Here $Im(f) = \{1, 2, 4\}$



Background on functions

Example:

Take $X = \{1, 2, 3, \dots, 10\}$ and $f: X \rightarrow X$ is defined as $f(x) = x * x \bmod 11$. We then have the following table:

x	1	2	3	4	5	6	7	8	9	10
f(x)	1	4	9	5	3	3	5	9	9	1

The image of f is $\text{Im}(f) = \{1, 3, 4, 5, 9\}$

One-to-one (1-1) Function:

A **function (transformation or mapping)** $f: X \rightarrow Y$ is 1-1 (injective) if each element in Y is the image of at most one element in X .

In other words,

$$x_1, x_2 \in X (x_1 \neq x_2)$$

$$\Rightarrow f(x_1) \neq f(x_2)$$

$$\text{i.e., } f(x_1) = f(x_2)$$

$$\Rightarrow x_1 = x_2$$

- **Onto (Surjective) function**

A function $f: X \rightarrow Y$ is onto if each element in Y is the image of at least one element in X . Equivalently, f is onto, if $\text{Im}(f) = Y$

- **Bijjective function (Bijection)**

If a function $f: X \rightarrow Y$ is 1-1 and onto, then it is called a bijection.

Theorem: If $f: X \rightarrow Y$ is 1-1, then $f: X \rightarrow \text{Im}(f)$ is a bijection.

Theorem: If $f: X \rightarrow Y$ is 1-1, and X and Y are finite sets of same sizes (cardinalities) then f is a bijection.

- Inverse Function: If a function $f: X \rightarrow Y$ is bijective, then its inverse exists.

If $g = f^{-1}$, then $g: Y \rightarrow X$.

Example:



- **One-Way Function**

A function $f: X \rightarrow Y$ is called a one-way function if $f(x)$ is “*easy*” to compute for all x in X but for “essentially all” elements y in $\text{Im}(f)$ it is “*computationally infeasible*” to find any x in X such that $f(x) = y$.

In other words, a one-way function is easily computed, but the calculation of its inverse is infeasible.

- **Trap-door one-way function**

A trap-door one-way function is a function $f: X \rightarrow Y$ with the additional property that given some extra information (called the trap-door information) it becomes feasible to find for any given y in $\text{Im}(f)$, an x in X such that $f(x) = y$.

In other words, we can say that a trap-door function is a function that is easily computed; the calculation of its inverse is infeasible unless privileged information is known

- **Permutation**

Let S be a finite set of elements. A permutation p on S is a bijection from S to itself (S). In other words, $p: S \rightarrow S$ is a bijection.

If $S = \{a_1, a_2, \dots, a_n\}$ then p is represented as

$$p = \begin{pmatrix} a_1 & a_2 & \dots & a_n \\ p(a_1) & p(a_2) & \dots & p(a_n) \end{pmatrix}$$

- **Inverse Permutation**

$$p^{-1} = \begin{pmatrix} p(a_1) & p(a_2) & \dots & p(a_n) \\ a_1 & a_2 & \dots & a_n \end{pmatrix}$$

Example: $S = \{1, 2, 3, 4\}$

$$p = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad p^{-1} = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix}$$

Basic terminology and concepts

- Encryption domains and co-domains

□ **A** denotes a finite set called the alphabet of definition.

For example, $A = \{0, 1\}$

□ **M** denotes a set called the *Message Space*. **M** consists of strings of symbols from an alphabet set of definition, **A**. An element of **M** is called a *plaintext message or simply a plaintext*.

□ **C** denotes a set called the *ciphertext space*. **C** consists of strings of symbols from an alphabet of definition, which may differ from the alphabet of definition. An element of **C** is called a *ciphertext*.

Encryption and Decryption Transformations

- K denotes a set called the key space. An element of K is called a key.
- Each element e in K uniquely determines a bijection function from M to C , denoted and defined by $E_e : M \rightarrow C$
 $m \rightarrow c$
is called an encryption function or encryption transformation.

- For each d in K , a decryption function or decryption transformation is defined as

$$D_d : C \rightarrow M$$
$$c \rightarrow m$$

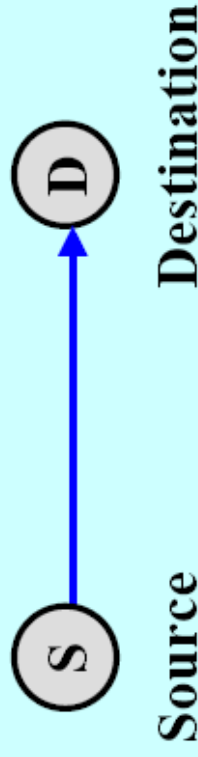
- An encryption scheme consists of a set $\{E_e : e \in K\}$ of encryption transformations and a corresponding set $\{D_d : d \in K\}$ of decryption transformations with the property that for each e in K there is a unique d in K such that

Requirements for constructing an encryption scheme

- A message space , M
- A ciphertext space, C
- A key space, K
- A set of encryption transformations $\{E_e : e \in K\}$
- A corresponding set of decryption transformations $\{D_d : d \in K\}$

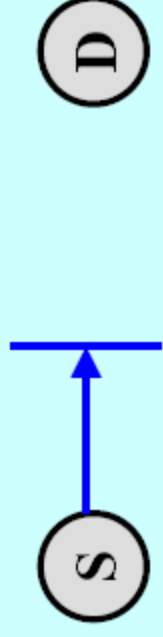
Security Attacks

- Any action that compromises the security of information.
- Four types of attack:
 1. Interruption
 2. Interception
 3. Modification
 4. Fabrication
- Basic model:

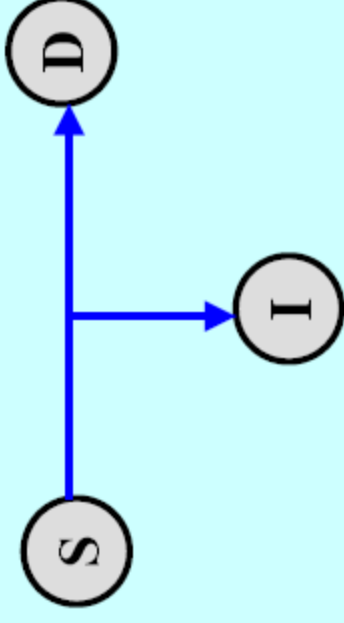


Security Attacks (Continued...)

- **Interruption:**
 - Attack on availability

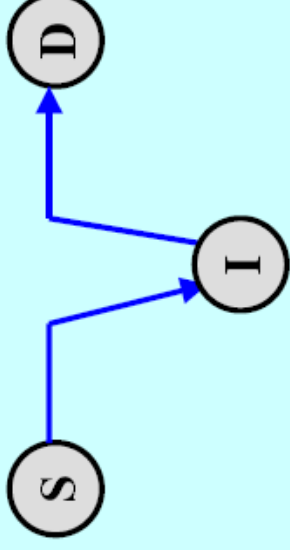


- **Interception:**
 - Attack on confidentiality

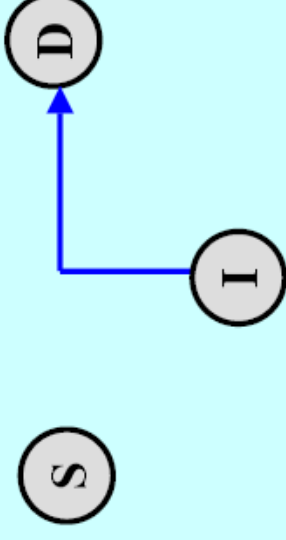


Security Attacks (Continued...)

- **Modification:**
 - Attack on integrity



- **Fabrication:**
 - Attack on authenticity



Passive and Active Attacks

- **Passive attacks**
 - Obtain information that is being transmitted (eavesdropping).
 - **Two types:**
 - Release of message contents:- It may be desirable to prevent the opponent from learning the contents of the transmission.
 - Traffic analysis:- The opponent can determine the location and identity of communicating hosts, and observe the frequency and length of messages being exchanged.
 - **Very difficult to detect.**

Passive and Active Attacks (Continued...)

- **Active attacks**
 - Involve some modification of the data stream or the creation of a false stream.
 - **Four categories:**
 - Masquerade:- One entity pretends to be a different entity.
 - Replay:- Passive capture of a data unit and its subsequent retransmission to produce an unauthorized effect.
 - Modification:- Some portion of a legitimate message is altered.
 - Denial of service:- Prevents the normal use of communication facilities.

Types of Attacks on Encrypted Messages

Type of Attack	Known to Cryptanalyst
Ciphertext only (most difficult)	•Encryption algorithm •Ciphertext to be decoded
Known Plaintext (easier than above)	•Encryption algorithm •Ciphertext to be decoded •One or more plaintext-ciphertext pairs formed with the secret key
Chosen plaintext (easier than above two)	•Encryption algorithm •Ciphertext to be decoded •Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key (*)
Chosen ciphertext (easier than above all)	•Encryption algorithm •Ciphertext to be decoded •Purported ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key (**)
Chosen text (easier than above all)	•Encryption algorithm •Ciphertext to be decoded •* •**
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Security

- **Unconditionally secure cipher**

An encryption scheme is “*unconditionally secure*” if the ciphertext generated by the scheme does not contain enough information to determine uniquely the corresponding plaintext, no matter how much ciphertext is available. In other words, no matter how much time an opponent has, it is impossible for him/her to decrypt the ciphertext, simply because the required information is not there.

- **Computationally secure cipher**

An encryption scheme is “*computationally secure*” if the following two conditions are met:

1. The cost of breaking the cipher exceeds the value of the encrypted information
2. The time required to break the cipher exceeds the useful lifetime of the information

Note: In practice, a scheme is considered as “computationally secure”, if the time needed to break the scheme is of $O(2^{80})$ operations.

Model of Network Security

